

spec_impl_alignment

Helix OTA — Spec-vs-Implementation Alignment Decision Memo (operational /api/v1)

Field	Value
Revision	5
Last modified	2026-06-10T14:50:00Z
Created	2026-06-08
Status	active (decision memo — additive + group breaking WIDENs landed; row-4 per-device telemetry filters + row-7 group/members pagination RESOLVED 2026-06-10; row-4 richer numeric fields duration_ms/bytes_transferred RESOLVED — now ingested end-to-end 2026-06-10) Row 4 richer fields RESOLVED — per-item duration_ms + bytes_transferred a now ingested end-to-end (71be1cd, feat(telemetry): per-event duration_ms + bytes_transferred end-to-end (now ingested)). The Tier-1 device emulator is now a real event source emitting duration_ms on every lifecycle event + bytes_transferred on download events; threaded additively emulator → server inger (handlers_client.go) → store (memory + pgx, nullable cols) → read VIEW (handlers_telemetry.go, omitempty). This closes the previously-DEFERRED “blocked on UNVERIFIED ingest (§9)” qualifier — the §9 register is updated to FACT below. Anti-bluff evidence: server/internal/deviceemu/telemetry_fields_test.go:TestEmulatorTelemetryFieldsEndToEnd asserts the EXACT values flow emulator→server→read-API over real HTTP; TestEmulatorTelemetryFieldsIngestMutation is the §1.1 paired mutation; real-Postgres parity proven via server/internal/store/postgres_integration_test.go:TestPostgresRepositoryContract_Integrat (podman-booted PG, 0 skips). Evidence dir docs/qa/20260610T140407Z-telemetry_fields/.
Landed (2026-06-10, telemetry numeric fields)	
Landed (2026-06-10)	Row 4 filters — per-device telemetry now accepts the spec filter params (50ef5c6, feat(api): per-device telemetry filters + group/members pagination; OpenAPI synced + redocly-clean). Row 7 pagination — group + members list now paginated. Both previously-DEFERRED WIDEN bits are RESOLVED. Dashboard API client updated in lockstep (b0b8ee2).
Operator ruling (2026-06-08)	WIDEN impl to the fuller spec for the remaining breaking divergences. Executing in sub-batches. Landed: group row 6 id-key id→group_id + row 8 member-add → batc {device_ids} → 200 {added, already_member, not_found} (device-registration-aware); row 1 audit actor → object {user_id, subject}; row 4 structural — per-

Field	Value
Landed (2026-06-08, operator- approved additive WIDENs)	device telemetry events→items + next_cursor, newest-first, ?limit/?cursor pagination (handler-side over bounded history). GET /groups/{id}/members → items[] of {device_id, added_at} (store gained an added_at membership column in ListGroupMembersDetailed on memory + pgx, real-DB parity). Remaining: row 4 DEFERRED bits (richer per-item fields duration_ms/bytes_transferred — UNVERIFIED ingest; ?event_type/?since/?until/?deployment_id filters), row 7 (group + members list pagination — deferred, groups bounded). Row 2 (audit query filters) — ?since/?until RFC3339 bounds on GET /audit (028e656). Row 5 core (telemetry overview) — failure_rate = failure/(success+failure) + by_state (028e656). Row 6 partial (member_count) — added to GroupView on get/list/update (live count via ListGroupMembers). All three additive (new optional params / new JSON fields), wire-compatible, real-DB parity where a store change applied. Remaining rows (1 audit actor object, 4 per-device telemetry pagination events→items, 8 batch member-add, plus the id→group_id rename) are BREAKING; wire changes still gated behind an operator WIDEN/TRIM ruling.
Status summary	A per-divergence decision memo over the eight spec-vs-implementation gaps recorded in <u>implemented_endpoints.md</u> §10 for the audit-read, telemetry-read, and device-group operational surfaces. For each gap it recommends WIDEN-impl (bring the handler up to the fuller <u>operational_endpoints.md</u> spec) or TRIM-spec (accept the leaner as-built shape as canonical and amend the prose spec), with rationale, effort, risk, and the blast radius (tests / e2e / HelixQA bank). It is a MEMO only — it recommends; it changes no server/ code, no go.mod, no tests, no existing file.
Authority	Recommendations are advisory. The operator decides WIDEN-vs-TRIM per row (Constitution §11.4.66 — closed-set decisions belong to the operator; §11.4.122 — no silent removal of a shipped capability).
Anti-bluff	Per Constitution §11.4.6 (no-guessing): every “impl does” cell was confirmed by direct read of the cited handler function in server/internal/api/; every “spec wants” cell was confirmed by direct read of <u>operational_endpoints.md</u>. The §10 register at the foot of this memo lists what is UNVERIFIED (effort/risk estimates, NFR numbers, store method availability) and is not asserted as fact.
Owner	Helix OTA control-plane / API team
Related	<u>implemented_endpoints.md</u> (as-built §10 gap list — source of this memo); <u>operational_endpoints.md</u> (fuller prose spec); <u>endpoints.md</u> (conventions/error-model/RBAC/pagination); <u>openapi.yaml</u>

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1. purpose_and_scope

implemented_endpoints.md §10 records eight concrete shape divergences between the prose operational spec (operational_endpoints.md) and the code that actually shipped (handlers_audit.go, handlers_telemetry.go, handlers_group.go). The as-built doc states the **code is authoritative** but does not decide which side *should* win going forward. This memo makes that decision per gap:

- **WIDEN-impl** — the spec's fuller shape carries real operational value; raise the handler to it (the as-built doc becomes a transient under-build state, the prose spec stays the target).
- **TRIM-spec** — the leaner as-built shape is sufficient for the 1.0.0-MVP; adopt it as canonical and amend operational_endpoints.md + openapi.yaml to match (no code churn; doc-sync only).

Scope is exactly the eight §10 gaps across audit / telemetry / groups. Out of scope: rollout (implemented_endpoints.md §10 marks it *consistent* with the brick), roles (also *consistent*), and the PLANNED recall endpoint (§8 — a missing feature, not a shape divergence).

2. how_to_read_this_memo

- **impl does cites file:func** — the exact handler confirmed by reading the source this pass.
- **WIDEN/TRIM** is the recommendation; the operator owns the final call (§11.4.66).
- **Effort** is a relative T-shirt size (S / M / L) for the handler+store+test work; the absolute hours are UNVERIFIED (§9) because the pgx store-method availability for the wider shapes is not confirmed in this memo.
- **Risk** is the chance the change breaks an existing green test or an as-built consumer.
- **Blast radius** names the test layers and HelixQA bank entries that would change. The HelixQA bank path (tools/helixqa/banks/atmosphere.yaml) is the constitutional convention (§11.4.58); whether that exact file exists in this repo is UNVERIFIED (§9) — treat it as “the project's Challenge bank, wherever it lives”.

3. evidence_basis_confirmed_against_code

Each divergence below was re-confirmed against source this pass (not taken on trust from §10):

- **Audit actor flat-string** — handlers_telemetry.go aside, the audit entry is built in handlers_audit.go:auditMiddleware (store.AuditEntry{ActorSubject: claims.Subject, ...}, L36–45) and rendered by toAuditLogEntry into AuditLogEntry; the wire actor is a single string, never a {user_id, subject} object. Spec operational_endpoints.md §4.3 shows "actor": { "user_id": ..., "subject": ... } (L234).

- **Audit query params** — `handlers_audit.go:handleListAudit` reads only `action`, `resource_type`, `cursor`, `limit` (L133–147). Spec §4.3 (L224–226) lists additionally `?actor`, `?resource_id`, `?since`, `?until`.
- **Audit response type name** — code emits `AuditLogList` (`handleListAudit`, L157); spec §4.3 calls it `AuditList` (L227).
- **Telemetry per-device** — `handlers_telemetry.go:handleDeviceTelemetry` consumes no query params, returns full history in store order under events (L51–67); per-event `TelemetryEventView` (L14–22) has no `id/success/duration_ms/bytes_transferred`. Spec §5.1 (L270–292) wants `?limit/?cursor/?event_type/?since/?until/?deployment_id`, newest-first, an items array, and those richer fields.
- **Telemetry overview** — `handlers_telemetry.go:handleTelemetryOverview` (L71–82) returns the minimal `TelemetryOverview{EventCounts, Total}` (L32–35), no scoping. Spec §5.2 (L308–342) wants `scope`, `devices_total`, `devices_reporting`, `by_state` (latest-per-device), `failure_rate`, `generated_at`, plus `?deployment_id/?os/?since/?until`.
- **Group view fields + id key** — `handlers_group.go:GroupView` (L28–33) is `{id, name, description, created_at}`; no `filter_criteria`, no `member_count`, id key is `id`. Spec §6.1 (L39–44) shows `group_id`, `filter_criteria`, `member_count`.
- **Group list pagination** — `handlers_group.go:handleListGroup`s (L79–90) returns all groups, `GroupList{Items}` (L36–38), no `?name/?limit/?cursor`, no `next_cursor`. Spec §6.2 (L53–54) wants paginated `{items, next_cursor}` + `?name=` prefix.
- **Group member add semantics** — `handlers_group.go:handleAddGroupMember` (L152–171) takes a single `MemberAdd{DeviceID}` (L41–43) and returns 204. Spec §6.4 (L89–98) wants a batch `{device_ids:[...]}` returning 200 with `GroupMemberAddResult{added, already_member, not_found}`.
- **Group member list shape** — `handlers_group.go:handleListGroupMembers` (L139–150) returns `GroupMembers{group_id, device_ids:[string]}`. Spec §6.4 (L76–85) wants paginated items of `{device_id, added_at}`.

4. decision_table

#	divergence	spec wants	impl does (file:func)
1	audit actor shape	nested {user_id, subject}	flat string = subject- (<code>handlers_audit.toAuditLogEntry</code>)
2	audit query filters	<code>?actor,?resource_id,?since,?until</code> (+ existing)	only <code>?action,?res</code> (<code>handlers_audit.</code>

#	divergence	spec wants	impl does (file:function)
3	audit response type name	AuditList	AuditLogList (handlers_audit.py)
4	per-device telemetry: pagination + newest-first + filters + richer fields	?limit/?cursor/?event_type/?since/?until/?deployment_id,newest-first, items[] with id/success/duration_ms/bytes_transferred	newest-first items [filter params RESOLVED — incoming (handlers_telemetryEventView.empty)
5	telemetry overview: richer fields + scoping	scope,devices_total,devices_reporting,by_state (latest-per-device),failure_rate,generated_at + ?deployment_id/?os/?since/?until	minimal {event_count} (handlers_telemetryOverview)

#	divergence	spec wants	impl does (file:function)
6	GroupView fields + id key	group_id, filter_criteria, member_count	id, no filter_cri (handlers_group.
7	GET /groups + GET /groups/ {id}/members pagination	paginated {items, next_cursor} (+ ?name= prefix on groups)	RESOLVED 2026- list now paginated (handlers_group. handleListGroupM
8	group member add: single+204 vs batch+200+result	batch {device_ids:[...]} → 200 {added, already_member, not_found}	single {device_id] (handlers_group. MemberAdd)

5. per_divergence_rationale

5.1 audit (rows 1–3)

The audit subsystem's only purpose is forensic reconstruction. Two of its three gaps directly defeat that purpose:

- **Row 1 (WIDEN).** The schema keeps a nullable `user_id` precisely so a record outlives the actor (`operational_endpoints.md` §4 intro, `ON DELETE SET NULL`). The handler currently flattens to a single subject string. Surfacing `{user_id, subject}` restores the durable join key for cross-actor queries; effort is small because both values already exist at write time (`claims.Subject` is captured; the `users.id` resolution the spec describes in §4.1 L140 is the only new lookup). **Effort S–M. Risk LOW** — additive JSON field; existing string consumers break only if they assumed actor is a string (the as-built doc says it is, so an as-built consumer would break — gate this behind the operator's WIDEN decision).
- **Row 2 (WIDEN).** `?since/?until` time-window scoping is the single most-used audit query in incident response; `?actor/?resource_id` complete the “who touched what” triad. The store seam (`store.AuditFilter`) is already the extension point — adding fields there + parsing+validating the RFC-3339 params in `handleListAudit` is the bulk of the work. **Effort M. Risk LOW-MED** — new validation paths (malformed `since/until` → 400) must be tested to avoid §11.4.1 FAIL-bluffs.
- **Row 3 (TRIM).** Type *name* only; the wire contract (`items, next_cursor`) is identical. Renaming `AuditLogList`→`AuditList` is pure churn. Adopt `AuditLogList` as canonical; fix the prose + `openapi`. **Effort S (doc only). Risk NONE.**

5.2 telemetry (rows 4–5)

- **Row 4 (WIDEN, phased).** Per-device history is unbounded and the handler returns *all of it in insertion order*. At any real device age this is a latency/payload problem and gives no “most-recent-first” affordance. Pagination + newest-first + `?event_type/?since/?until` filtering are load-bearing and should be widened. The richer fields split: `id` and `success` are cheap derivations; `duration_ms/bytes_transferred` require the telemetry ingest to *carry* those values — whether `store.TelemetryRecord` / the ota-protocol event already has them is **UNVERIFIED (§9)**. Phase: ship pagination+order+filters first; add the rich numeric fields only once the ingest is confirmed to supply them (else they would be a §11.4-bluff of always-zero fields). **Effort M–L. Risk MED** — touches the store query contract (`TelemetryForDevice` gains filter/cursor params) and changes the response *key* (`events`→`items`), which is a breaking wire change an as-built consumer would feel; gate behind the operator decision.
- **Row 5 (WIDEN core / TRIM ?os).** `failure_rate + by_state` computed as *latest-event-per-device* is the honest fleet-health number; the current raw `event_counts` over-weights chatty devices and has no health ratio at all. Widen to `by_state/failure_rate/devices_total/devices_reporting/generated_at`. The latest-per-device aggregate is a `DISTINCT ON (device_id)` query (spec §5.3 alludes to it) — real work in the store. `?os` scoping is the weakest item (needs a device-OS join, low MVP demand) — TRIM it for 1.0.0, keep `?deployment_id` + time window. **Effort L. Risk MED**

— additive response fields are safe; the new store aggregate must be tested for the zero-devices edge (`failure_rate = 0` when `devices_reporting = 0`, per spec §5.2 L345).

5.3 groups (rows 6–8)

- **Row 6 (split).** Three sub-decisions: **id key id** (TRIM — matches the rest of the API, renaming to `group_id` churns every group test for cosmetics); **member_count** (WIDEN — a cheap COUNT, high-value in list/detail UIs); **filter_criteria** (TRIM for MVP — the spec *itself* flags dynamic-membership evaluation as UNVERIFIED and MVP as static-only, §6 intro L13–14; storing+returning an un-evaluated selector is a latent §11.4-bluff “feature that doesn’t work”, so defer until the evaluator lands). **Effort S–M. Risk LOW.**
- **Row 7 (TRIM, defer).** Groups are operator-curated and bounded (not per-device rows), so an un-paginated list is acceptable at MVP scale. Adopt the lean shape as canonical now; document the paginated `{items, next_cursor} + ?name=` as a future widen triggered if groups ever get created programmatically. **Effort S (doc). Risk LOW.**
- **Row 8 (WIDEN).** Membership is inherently bulk (build a cohort = add many devices). Single-add forces N round-trips + N audit rows and gives no partial-success feedback; the spec’s `{added, already_member, not_found}` is precisely the report an operator needs, and the store method is plural by design (AddGroupMembers). The `204→200+body` and `device_id→device_ids` changes are breaking for an as-built single-add consumer — gate behind the operator decision. **Effort M. Risk MED** — must test idempotent re-add (counted, not error) and unknown-device (`not_found`, partial success) to avoid a §11.4.1 FAIL-bluff on the partial path.

6. overall_recommendation_first_things_first

Order by (operational-harm-if-left-lean) × (low effort/risk first), and honor §11.4.72 only where applicable (no audio surface here, so pure operational priority):

1. **Audit query filters (row 2) — FIRST.** Highest harm: without `?since/?until/?actor/?resource_id` the audit trail cannot answer the incident question it exists for. Medium effort, low-med risk, purely additive query params (no breaking wire change). Biggest forensic value per unit work.
2. **Telemetry overview core fields (row 5, minus ?os) — SECOND.** `failure_rate + by_state` is the fleet-health number operators watch during a rollout; additive response fields are non-breaking. Pairs naturally with the rollout/telemetry story.
3. **Per-device telemetry pagination + newest-first + filters + rich fields (row 4) — FULLY LANDED 2026-06-10.** Prevented the unbounded-history payload cliff (pagination + newest-first + filters, 50ef5c6). The `duration_ms/bytes_transferred` fields — previously deferred until the ingest was confirmed to supply them — now ingest end-to-end (71be1cd): the Tier-1 emulator is the real event source. The breaking key change (`events→items`) landed with the dashboard client updated in lockstep. Row 4 is no longer a standing divergence.
4. **Group member batch-add (row 8) — FOURTH.** Real ergonomics win for cohort building; breaking shape change so it rides with row 4 under the operator decision.

5. **Audit actor object (row 1) — FIFTH.** Restores the durable user_id join key; additive but needs the users resolution lookup; lower urgency than time-window filtering.
6. **member_count on GroupView (row 6 partial) — SIXTH.** Cheap UI affordance; do it alongside any group-handler touch.
7. **TRIM-spec items (rows 3, 7, and the id-key + filter_criteria + ?os sub-parts of 5/6) — ANYTIME, doc-only.** No code; amend operational_endpoints.md + openapi.yaml so the prose stops disagreeing with the shipped code (§11.4.6 — divergences must not sit silently). Land these as a single doc-sync change because they cost nothing and immediately stop the spec from lying.

Net: do the three TRIM doc-syncs immediately (zero code, removes the standing divergence), then work the WIDEN list in the 1→6 order above behind a single operator WIDEN/TRIM ruling per breaking row (§11.4.66). Rows 1, 2, 5 are additive/non-breaking and can land without consumer coordination; rows 4, 8 (and the actor-object visibility change in 1) are breaking wire changes that need the as-built consumer sweep first.

7. blast_radius_summary

layer	rows that touch it	what changes
wire-struct unit tests	1,3,4,5,6,8	JSON tag / field-set assertions for AuditLogEntry, TelemetryEventView, TelemetryOverview, GroupView, MemberAdd→batch, AuditLogList/AuditList name
handler unit/integration tests	1,2,4,5,8	query-param parsing + validation (400 on bad since/until/limit), newest-first ordering, batch idempotency + not_found partial success, aggregate edge (failure_rate at 0 devices)
store seam tests	1,2,4,5,8	AuditFilter (actor/resource/time), TelemetryForDevice (filter+cursor), new latest-per-device overview aggregate, AddGroupMember→AddGroupMembers, member_count query
e2e / HelixQA Challenge bank	1,2,4,5,6,8	audit-filter, fleet-health, device-history, group-CRUD, group-membership Challenges each re-assert the new shape with captured evidence (§11.4.69 / §11.4.83)
openapi.yaml	all 8	path/schema definitions re-aligned to whichever side wins each row
audit middleware	8	none — deriveAuditAction already emits group_member verbs for the members sub-routes

layer	rows that touch it	what changes
		(handlers_audit.go), so batch-add stays correctly audited

8. doc_sync_actions_if_trim_chosen

For every row the operator rules TRIM-spec, the prose spec and openapi MUST be amended in the same change so they stop disagreeing with the shipped code (Constitution §11.4.6 — no silent divergence; §11.4.65 — keep .md/.html/.pdf siblings in sync):

- Row 3 → operational_endpoints.md §4.3: rename response model AuditList → AuditLogList.
- Row 6 (id) → §6.1/§6.2: change group_id → id in the Group/GroupView examples.
- Row 6 (filter_criteria) → §6.1/§6.3: mark filter_criteria as “stored, MVP returns it but evaluation is deferred — static membership only” (the spec already half-says this at §6 intro).
- Row 7 → §6.2: note GET /groups is un-paginated in 1.0.0 (lean shape canonical); keep the paginated form as a documented future-widen.
- Row 5 (?os) → §5.2: drop ?os from the 1.0.0 scope set (keep ?deployment_id + time window).

This memo does **not** perform those edits — it recommends them pending the operator’s per-row ruling.

9. anti_bluff_unverified_register

Per Constitution §11.4.6, the following are explicitly **not asserted as fact** in this memo:

- **UNVERIFIED — effort/risk sizes.** The S/M/L sizes and LOW/MED risk labels are estimates from reading the handlers + store seam, not from a spiked implementation. Absolute hours are not claimed.
- **RESOLVED — telemetry rich fields availability (row 4) (was UNVERIFIED; now FACT 2026-06-10, 71be1cd).** The earlier qualifier (“whether store.TelemetryRecord / the ota-protocol telemetry event already carries duration_ms / bytes_transferred”) is no longer open: the Tier-1 device emulator was made a real event source that emits duration_ms on every lifecycle event + bytes_transferred on download events, threaded additively emulator → server ingest (handlers_client.go) → store (memory + pgx, nullable columns) → read VIEW (handlers_telemetry.go TelemetryEventView, omitempty). The “always-zero columns (§11.4-bluff)” risk is closed because the values are ingested end-to-end and asserted by exact-value tests — NOT a zero default. Proof: server/internal/deviceemu/telemetry_fields_test.go:TestEmulatorTelemetryFieldsEndToEnd
 - TestEmulatorTelemetryFieldsIngestMutation (§1.1 pair) + real-PG parity server/internal/store/postgres_integration_test.go:TestPostgresRepositoryContract

_Integration; evidence docs/qa/20260610T140407Z-telemetry-fields/.

- **UNVERIFIED — store-method availability for wider shapes.** The wider audit filter, paginated telemetry query, latest-per-device overview aggregate (DISTINCT ON), member_count count, and AddGroupMembers plural method are named by the spec (operational_endpoints.md §4.2/§5.3/§6.5) but their *existence in the pgx/in-memory store.Repository* was not verified here. Effort rises if these must be added to the store.
- **UNVERIFIED — HelixQA bank path.** tools/helixqa/banks/atmosphere.yaml is the constitutional convention (§11.4.58); its presence in *this* repo was not confirmed. “Bank” in §4/§7 means the project’s Challenge bank wherever it lives.
- **UNVERIFIED — as-built external consumers.** The “breaking wire change” risk on rows 1/4/8 assumes a consumer that already depends on the lean as-built shape. Whether such a consumer (dashboard, device agent) exists and is pinned to the lean shape was not confirmed; the breaking qualifier is a conservative §11.4.6 default (treat as breaking unless proven otherwise), and is exactly why those rows are gated behind an operator §11.4.66 decision and a consumer sweep.
- **FACT (confirmed this pass).** Every “impl does” cell and its file:func citation, and every “spec wants” cell, were read directly from the cited source files in server/internal/api/ and operational_endpoints.md during this memo’s preparation (see §3).